

Science Olympiad — State Competition:

Codebusters C

2019

Names of participants: (Please print neatly)

School Name:

V JV1 JV2 JV3

Warning: Do not open this packet until given permission to do so.

Note: There are useful notes after this page.

Scoring:

Time to solve first problem: _____ (use to calculate Bonus below)

Question	Value	Incorrect letters	Deduction	Score
Timed	400			
1	300			
2	250			
3	250			
4	250			
5	200			
6	100			
7	350			
8	250			
9	100			
10	150			
11	200			
12	200			
13	200			
14	150			
15	300			
16	250			
17	250			
18	300			
Bonus				
Final Score				

The following tables might be useful during the event.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

1	3	5	7	9	11	15	17	19	21	23	25
1	9	21	15	3	19	7	23	11	5	17	25

AAAAA	A	AABBA	G	ABBAA	N	BAABA	T
AAAAB	B	AABBB	H	ABBAB	O	BAABB	U/V
AAABA	C	ABAAA	I/J	ABBBA	P	BABAA	W
AAABB	D	ABAAB	K	ABBBB	Q	BABAB	X
AABAA	E	ABABA	L	BAAAA	R	BABBA	Y
AABAB	F	ABABB	M	BAAAB	S	BABBB	Z

Frequency Table of English letters:

E - 12.51%	S - 6.54%	C - 3.06%	G - 1.96%	K - 0.67%
T - 9.25%	R - 6.12%	U - 2.71%	W - 1.92%	X - 0.19%
A - 8.04%	H - 5.49%	M - 2.53%	Y - 1.73%	J - 0.16%
O - 7.60%	L - 4.14%	F - 2.30%	B - 1.54%	Q - 0.11%
I - 7.26%	D - 3.99%	P - 2.00%	V - 0.99%	Z - 0.09%
N - 7.09%				

Frequency Table of Spanish letters:

E - 14.08%	I - 5.98%	M - 3.08%	Y - 1.09%	Z - 0.47%
A - 12.16%	L - 5.24%	P - 2.89%	V - 1.05%	Ñ - 0.17%
O - 9.20%	D - 4.67%	B - 1.49%	G - 1.00%	X - 0.14%
S - 7.20%	T - 4.60%	H - 1.18%	F - 0.69%	K - 0.11%
N - 6.83%	U - 4.69%	Q - 1.11%	J - 0.52%	W - 0.04%
R - 6.41%	C - 3.87%			

For the purposes of cryptograms it is customary to treat n and ñ as distinct letters, but a and á are the same letter. Likewise for e and é, and i and í. In other words, all the accent marks get amputated when working with cryptograms. Also, while some older Spanish dictionaries consider ch, ll, and rr, to be their own letters—this has fallen out of modern usage. Accordingly, “burro” is considered as five letters: “b-u-r-r-o” and not as four letters “b-u-rr-o.”

Famous Phrases

Gettysburg address

FOUR SCORE AND SEVEN YEARS AGO OUR FATHERS BROUGHT FORTH ON THIS CONTINENT, A NEW NATION, CONCEIVED IN LIBERTY, AND DEDICATED TO THE PROPOSITION THAT ALL MEN ARE CREATED EQUAL.

Declaration of Independence

WHEN IN THE COURSE OF HUMAN EVENTS IT BECOMES NECESSARY FOR ONE PEOPLE TO DISSOLVE THE POLITICAL BANDS WHICH HAVE CONNECTED THEM WITH ANOTHER AND TO ASSUME AMONG THE POWERS OF THE EARTH, THE SEPARATE AND EQUAL STATION TO WHICH THE LAWS OF NATURE AND OF NATURE'S GOD ENTITLE THEM, A DECENT RESPECT TO THE OPINIONS OF MANKIND REQUIRES THAT THEY SHOULD DECLARE THE CAUSES WHICH IMPEL THEM TO THE SEPARATION.

Constitution of United States of America

WE THE PEOPLE OF THE UNITED STATES, IN ORDER TO FORM A MORE PERFECT UNION, ESTABLISH JUSTICE, INSURE DOMESTIC TRANQUILITY, PROVIDE FOR THE COMMON DEFENSE, PROMOTE THE GENERAL WELFARE, AND SECURE THE BLESSINGS OF LIBERTY TO OURSELVES AND OUR POSTERITY, DO ORDAIN AND ESTABLISH THIS CONSTITUTION FOR THE UNITED STATES OF AMERICA.

MAGNA CARTA (In Latin)

JOHANNES DEI GRACIA REX ANGLIE, DOMINUS HIBERNIE, DUX NORMANNIE, AQUITANNIE ET COMES ANDEGAVIE, ARCHIEPISCOPI, EPISCOPI, ABBATIBUS, COMITIBUS, BARONIBUS, JUSTICIARIIS, FORESTARIIS, VICECOMITIBUS, PREPOSITIS, MINISTRIS ET OMNIBUS BALLIVIS ET FIDELIBUS SUIS SALUTEM.

Timed Question **[400 points]** Decode this Aristocrat which is a message from your event supervisor. When you have solved it, raise your hand so that the time can be recorded and the solution checked.

HSWFNYEPRYEZSWT JSN QYCZWF ZE ES EMB TEYEB

ESPNWYQBWE! FSSI RPHC UZEM KSPN BOBWET YWI MYOB Y

FNBYE IYK.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency		7	2		12	4		2	3	1	2		3	5	2	4	2	2	8	3	1		7		10	4
Replacement																										

1) [300 points] Decode this Harper Lee quote with a Baconian Cipher

actatcatcataccctcatatccatactacctactatcccatcatcatatato

atcatactcatactaccctatccatcatccactccatatacatactactcatat

ccatatacctccactactcatcactccactcactactacctatcatcactcca

ctcactactacctato

2) [250 points] Solve this K1 encoded Xenocrypt which is a translation of a quote by Paul Valéry into Spanish.

IBHBÑPT CCEÑES GLBFGLE TPCP EC GPFMVUP IB CET

JPSÑVCET RVB USLVFJEF TLBÑQSB. UPIP BC SBTUP BT

CLUBSEUVSE.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Ñ	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency		11	8		9	5	3	1	3	2		5	1		4		9	1	1	7	7	6	5				
Replacement																											

3) [250 points] Decode this Xenocrypt which is the infinitive version of a Spanish idiom.

BIV DKWK RFBDHV FAH HEFZH IA FA THZHV.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Ñ	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	2		2	1	4		6	2		2								1		1		3	1			2
Replacement																											

4) [250 points] Decode this inspirational Winston Churchill quote that has been encoded using a Running Key Cipher against a famous document.

O	Y	V	J	I	H	W																					

W	H																										

Y	S	H																									

K	B	U	E	F																							

S	I	B	P	X	J	X																					

I	L																										

R	G	B																									

S	O	K	D	P																							

Z	M																										

W	X																										

H	Y	Q																									

C	A	I	I	E	V	I																					

K	T																										

G	Q	G	N	V	V	I	R																				

X	Z	T	T																								

D	Z	C	F	A	B																						

5) [200 points] Encode the Zeroth Law of Thermodynamics using the Vigenère Cipher with the key word "CLAUSIUS"

I	F		

T	W	O	

S	Y	S	T	E	M	S

A	R	E	

E	A	C	H	

I	N	

T	H	E	R	M	A	L

E	Q	U	I	L	I	B	R	I	U	M

W	I	T	H	

A	

T	H	I	R	D	

S	Y	S	T	E	M	,

T	H	E	Y

A	R	E	

I	N	

T	H	E	R	M	A	L

E	Q	U	I	L	I	B	R	I	U	M

W	I	T	H	

E	A	C	H	

O	T	H	E	R	.

T	H	I	S	

L	A	W	

H	E	L	P	S	

D	E	F	I	N	E

T	H	E	

C	O	N	C	E	P	T

O	F	

T	E	M	P	E	R	A	T	U	R	E	.

6) [100 points] Using a key of **FLUX**, compute the decryption matrix for a 2x2 Hill cipher using a 26 character alphabet.

$$\begin{pmatrix} F & L \\ U & X \end{pmatrix} \equiv \begin{pmatrix} 5 & 11 \\ 20 & 23 \end{pmatrix}$$

$$\begin{pmatrix} \boxed{} & \boxed{} \\ \boxed{} & \boxed{} \end{pmatrix}$$

9) [100 points] The name of this famous cryptanalyst is encoded with a Baconian Cipher. Decode this man's full name.

AAAAAABABAAAAAABBAAABABBAAAAABAABAAABBBABAAABAAABABB

ABABBAABAABABAABBBAAAAABAAAABBAABBA

10) [150 points] Decode this John Green quote with a Caesar Cipher.

N	Y	R	K	Z	J	K	Y	V	G	F	Z	E	K	F	W	S	V	Z	E	X	R	C	Z	M	V	Z	W	P	F	L

U	F	E	,	K	R	K	C	V	R	J	K	K	I	P	K	F	U	F	J	F	D	V	K	Y	Z	E	X

I	V	D	R	I	B	R	S	C	V	?
										?

11) [200 points] Encode this famous quote from Hamlet using the Affine Cipher with $a = 7$ and $b = 11$.

T	H	O	U	G	H	T	H	I	S	B	E	M	A	D	N	E	S	S	,	Y	E	T	T	H	E	R	E	I	S

M	E	T	H	O	D	I	N	,	T	.

12) [200 points] Decode this lyric written by Alan Jay Lerner which has been encoded using the Atbash Cipher

G	S	V	I	Z	R	M	R	M	H	K	Z	R	M	H	G	Z	B	H	N	Z	R	M	O	B	R	M	G	S	V

K	O	Z	R	M

13) **[200 points]** Andrew and Joshua want to communicate with each other using RSA for encryption. Andrew generates RSA keys obtaining the following values:

$$\begin{array}{ll} \Phi = 142080 & q = 241 \\ e = 51361 & d = 117601 \\ p = 593 & n = 142913 \end{array}$$

Likewise, Joshua also generates RSA keys resulting in the values

$$\begin{array}{ll} p = 359 & \Phi = 141768 \\ n = 142523 & q = 397 \\ e = 78599 & d = 107231 \end{array}$$

They ask each other for the public keys in order to communicate. What information do they each need to transmit in response?

You must also determine what formula Andrew needs to calculate in order to decrypt the value 66026 from Joshua

Enter the minimum values that Andrew needs to transmit to Joshua:

--	--	--

Enter the minimum values that Joshua needs to transmit to Andrew:

--	--	--

Write the formula Andrew needs to calculate in order to decrypt the value 66026 from Joshua

--

14) **[150 points]** Decode this quote from our 13th president, Millard Fillmore using the Vigenère Cipher with the key **PRESIDENT**.

P	E

L	G	V	R	V	N	U	A	V

H	W	N	H	E	G

B	H

S	I	L	B	H	V

G	A	P	E

E

V	Q	V	L	B	G	D	I	E	T	T	H	Z	V	V	I	F	V	Q

15) [300 points] This quote from Ralph Waldo Emerson has been encoded using the Running Key Cipher against a famous document. What does it say?

Y	V	Y	Y	W	C	Z	K	L	O	S	W	Z	I	Z	C	R	Q	I	E	D	K	T	U	R	S	G	R	S	D	T

O	S	I	A	A	F	B	.	Q	K	H	K	J	B	V	O	L	F	G	B	Y	M	V	,	U	C	Y	H	V	T	E	F
							.																,								

P	E	P	E	J	F	E	X	N	O	E	G	B	B	W	K	P	.
																	.

16) [250 points] Using a 3x3 Hill Cipher with a 26 character alphabet and a key of **TELEPHONE**, encode this Franklin D. Roosevelt quote.

$$\begin{pmatrix} T & E & L \\ E & P & H \\ O & N & E \end{pmatrix} \equiv \begin{pmatrix} 19 & 4 & 11 \\ 4 & 15 & 7 \\ 14 & 13 & 4 \end{pmatrix}$$

A	S	M	O	O	T	H	S	E	A	N	E	V	E	R	M	A	D	E	A	S	K	I	L

17) **[250 points]** Decode the name of a famous scientist using the Hill Cipher and the key **HARP**.

$$\begin{pmatrix} H & A \\ R & P \end{pmatrix} \equiv \begin{pmatrix} 7 & 0 \\ 17 & 15 \end{pmatrix}$$

L	X	G	E	W	Y	Z	N	P	X	G	W	F	T	C	Z	Z	Q

18) **[300 points]** Mikaela and Lauren are accountants for a very large bank, and have started a friendship. They communicate via email, because they live thousands of miles apart. Lauren gets curious and asks Mikaela the year that they were born. Mikaela doesn't mind telling Lauren, but they know that the bank monitors all employee emails, and is afraid of being the victim of age discrimination. Therefore, Lauren suggests that they use RSA, and they provides their public key: (5767, 3349). Mikaela replies with the ciphertext 873. Lauren's private key is 109. In what year was Mikaela born?

Enter the answer: